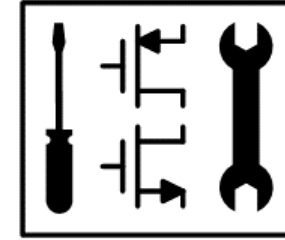


وَمَا أُوتِيتُمْ مِنَ الْعِلْمِ إِلَّا قَلِيلًا



Presents



Capacitive Feedback Amplifier

Dr. Hesham Omran

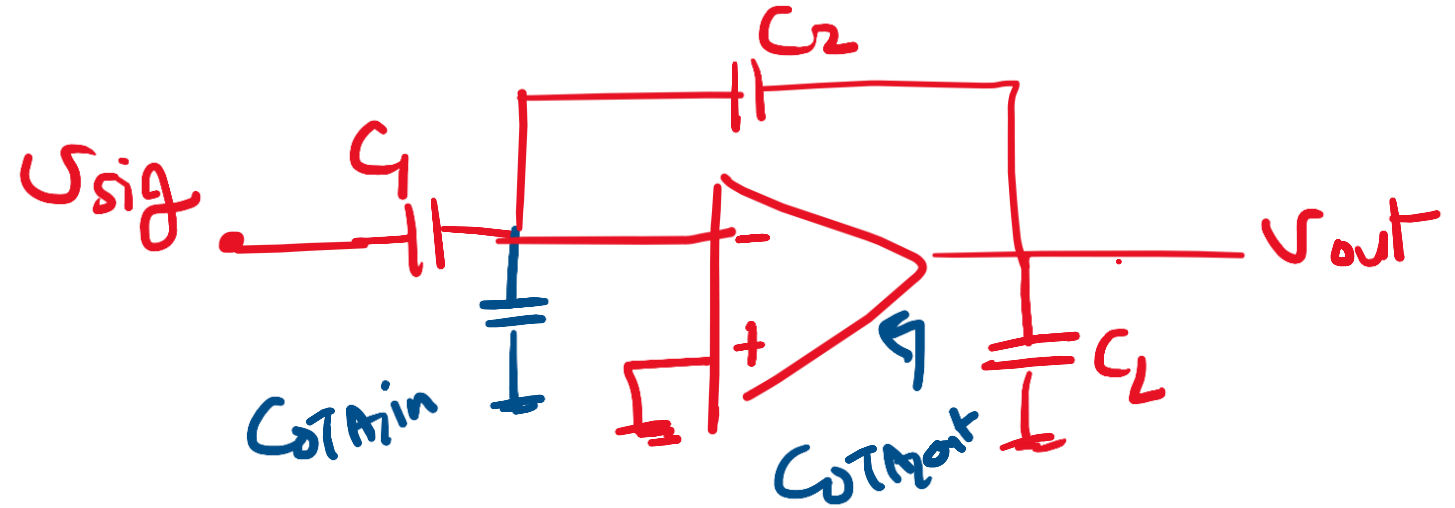
Master Micro LLC and Ain Shams University

adt.master-micro.com

Inverting vs Non-inverting

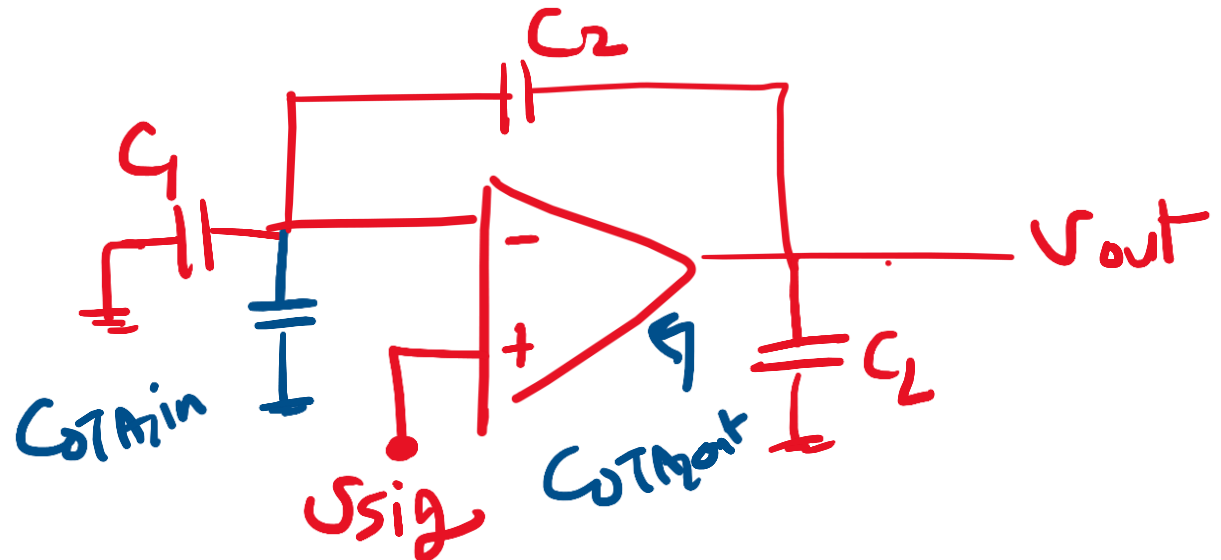
□ Inverting

- Input cap = C_1
- $A_{vCL} \approx -C_1/C_2$
- V_{inCM} is fixed
- Shunt mixing – shunt sampling
- Essentially a transimpedance amp



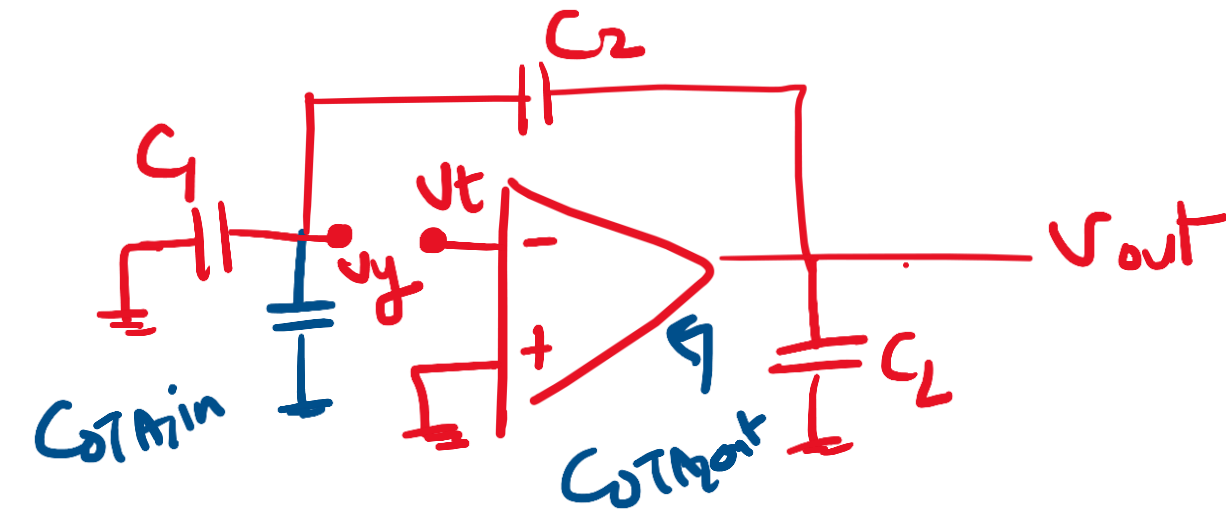
□ Non-inverting

- Input cap = $C_{OTA,in}$
- $A_{vCL} \approx 1 + (C_1 + C_{OTA,in})/C_2$
- V_{inCM} is varying
- Series mixing – shunt sampling
- A voltage amplifier



Loop Gain Analysis

- ❑ LG is the same for both inverting and non-inverting topologies.
- ❑ $C_{OTA,in}$ is frequency dependent in general (see [this](#)).
- ❑ The LG analysis here doesn't involve the "short shunt and sever series" approximation.



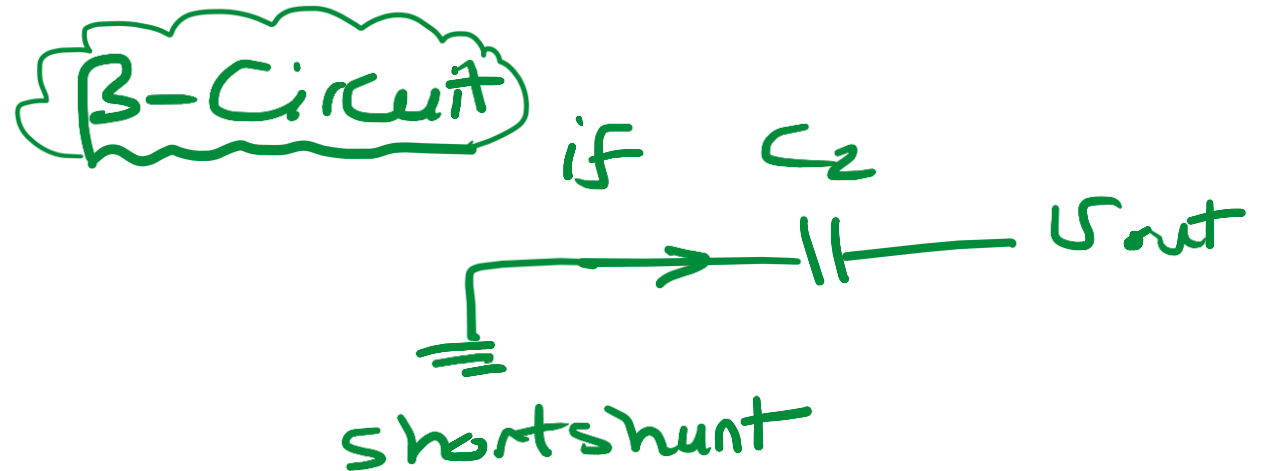
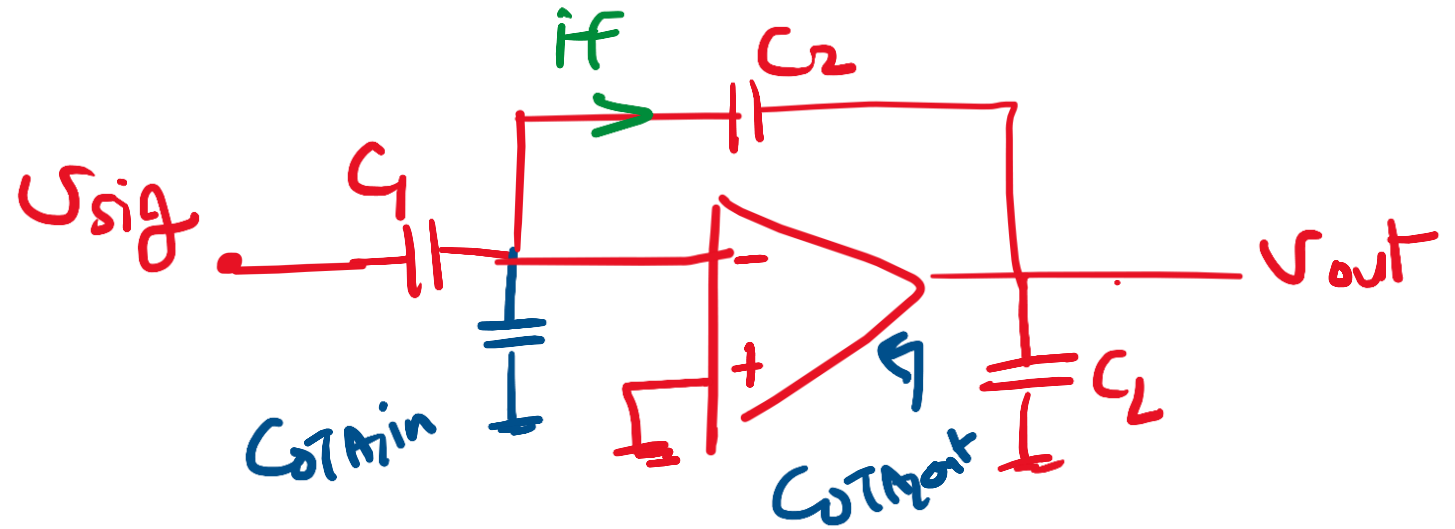
$$LG = - \frac{V_y}{V_t} = \frac{A_o}{1 + \frac{s}{\omega_{P1}}} \cdot \beta$$

$$C_{Leff} = C_L + C_{OTA,out} + C_2 \parallel (C_1 + C_{OTA,in})$$

$$\beta = \frac{C_2}{C_2 + C_1 + C_{OTA,in}}$$

Inverting (Shunt-Shunt) Analysis

$$\beta_G = \frac{i_f}{v_{out}} = -sC_2$$



Inverting (Shunt-Shunt) Analysis

$$i_{sig} = U_{sig}(s C_1)$$

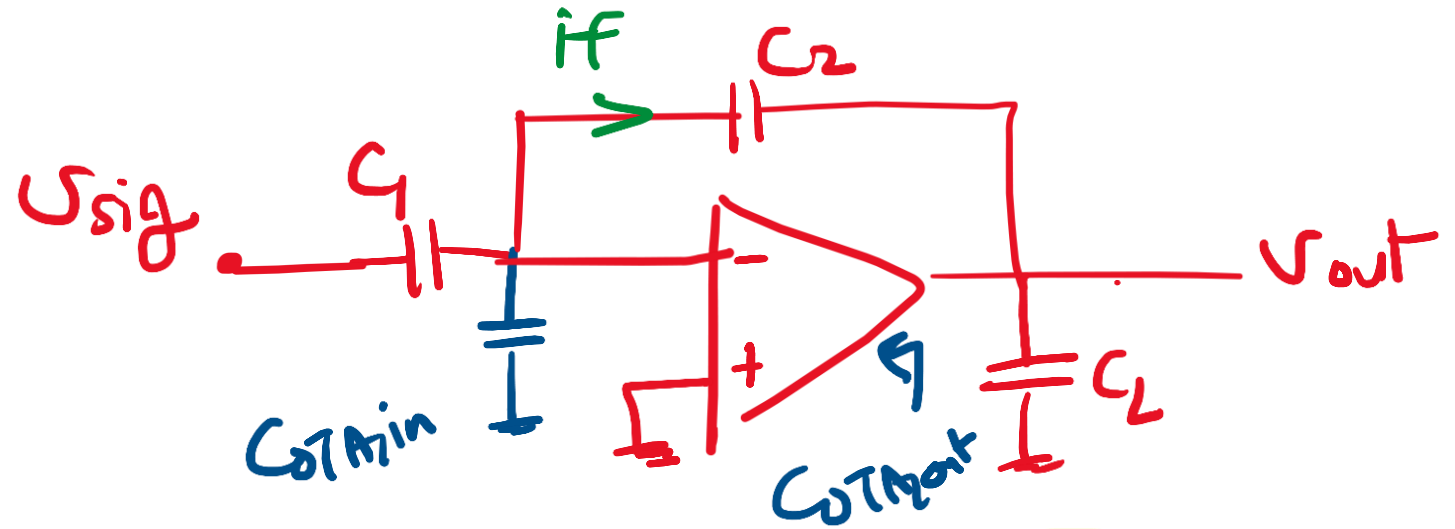
$$A_{R_{OL}} = \frac{U_{out}}{i_{sig}} = \frac{U_{in} U_{out}}{i_{sig} U_{in}}$$

$$A_{R_{OL}} = \frac{-A_{OL}(s)}{s(C_1 + C_2 + C_{OTAin})}$$

$$LG = \beta_G A_{R_{OL}}$$

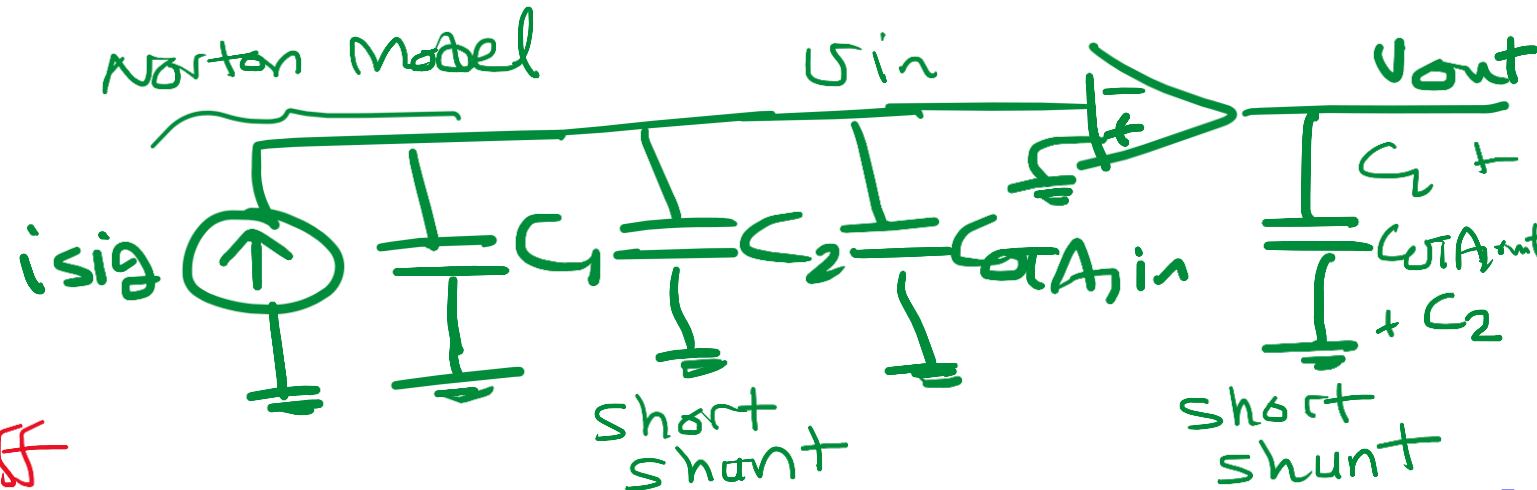
$$= \frac{C_2}{C_1 + C_2 + C_{OTAin}} \times A_{OL}(s)$$

depends on C_{Leff}



A Circuit

C_{Leff} is approx here



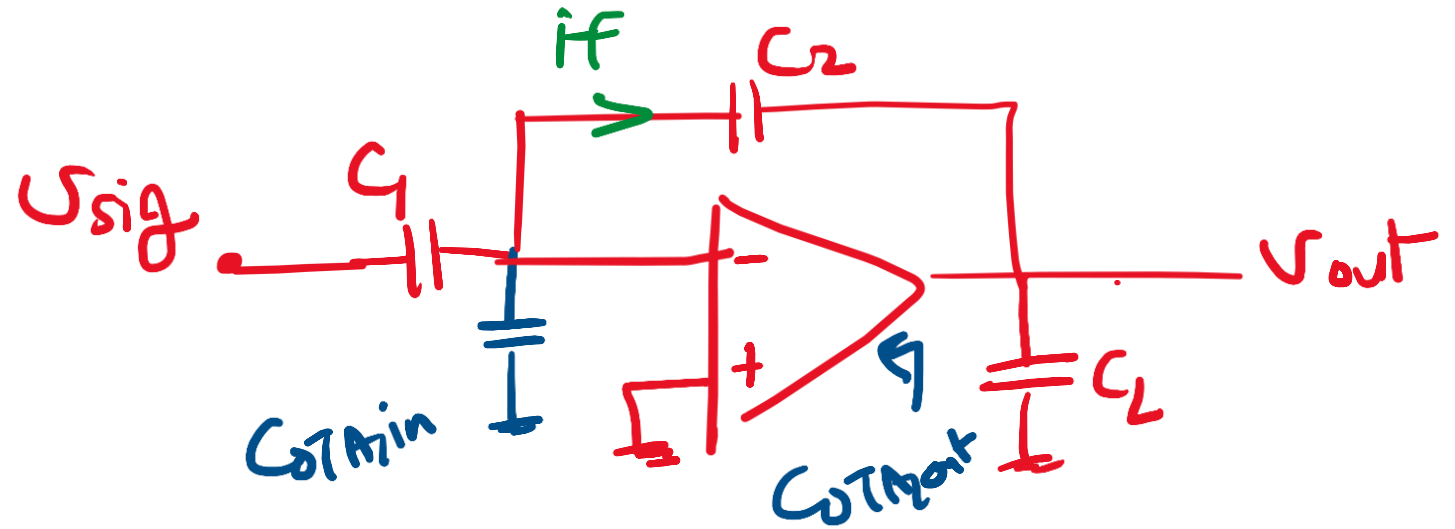
Inverting (Shunt-Shunt) Analysis

$$A_{R,C_2} = \frac{v_{out}}{i_{sig}}$$

$$\approx \frac{1}{\beta_G} = -\frac{1}{sC_2}$$

$$\frac{v_{out}}{v_{sig}} = \frac{v_{out}}{i_{sig}} \cdot \frac{i_{sig}}{v_{sig}}$$

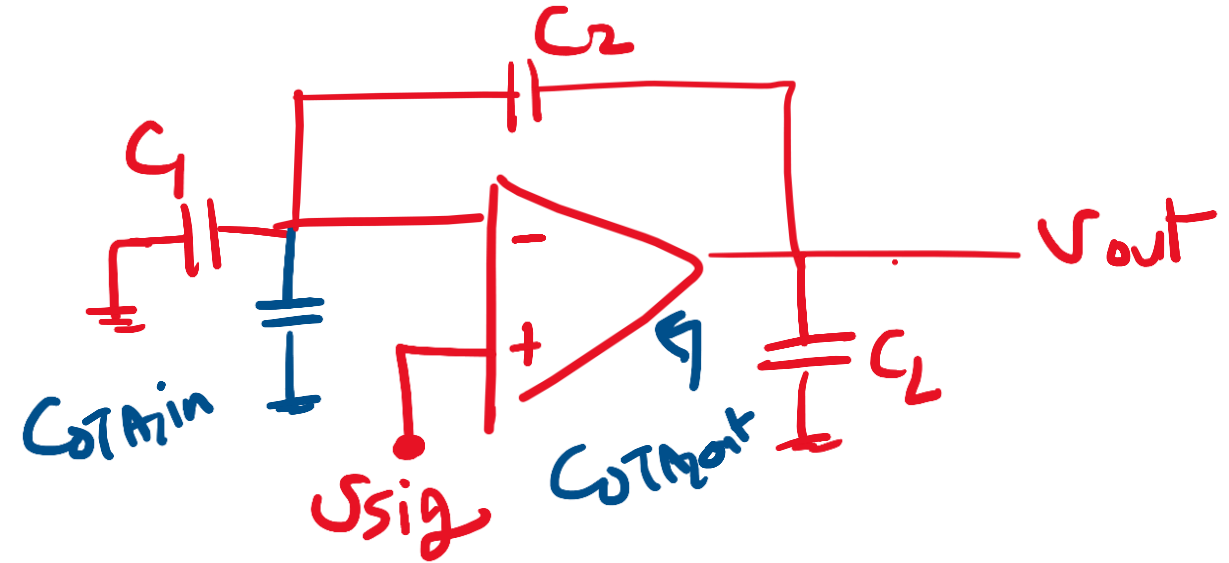
$$\approx -\frac{C_1}{C_2} \#$$



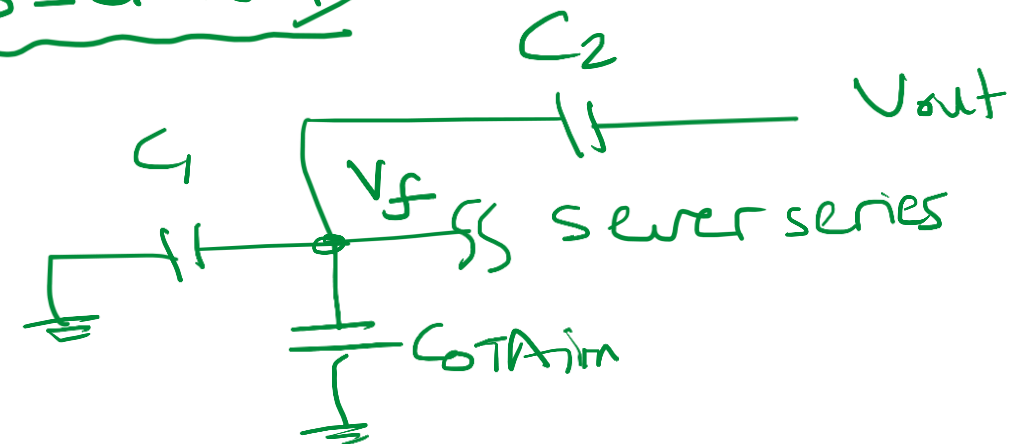
Non-Inverting (Series-Shunt) Analysis

$$\beta_v = \frac{V_f}{V_{out}}$$

$$= \frac{C_2}{C_1 + C_2 + C_{OTAin}}$$



β -Circuit



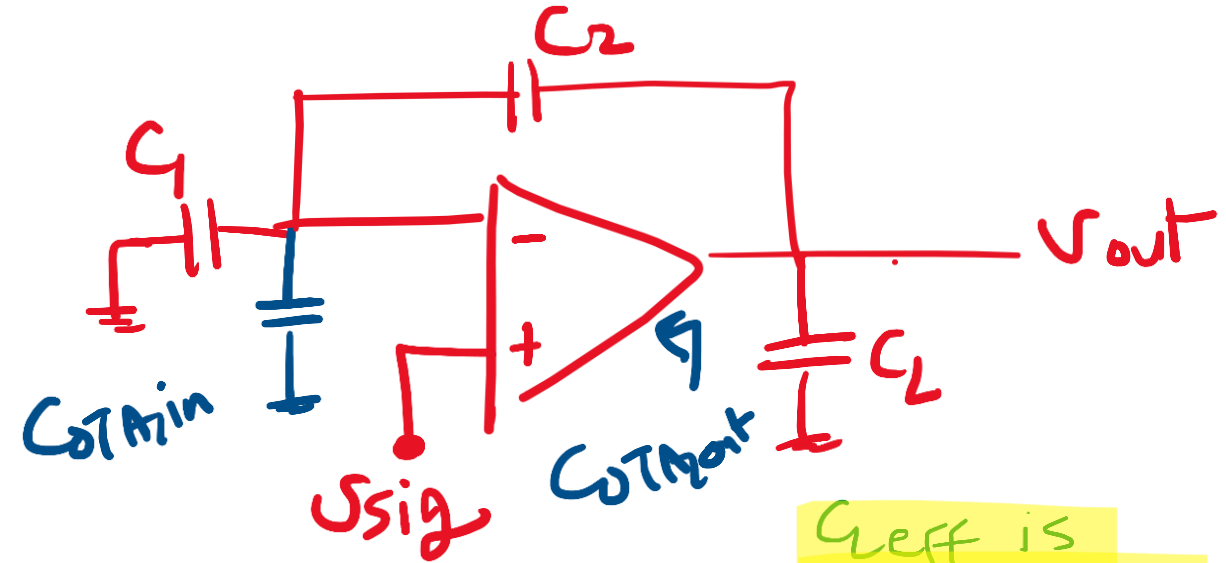
Non-Inverting (Series-Shunt) Analysis

$$A_{v,OL} = A_{OL}(s)$$

$$LG = \beta_v A_{v,OL}$$

$$= \frac{C_2}{C_1 + C_2 + C_{OTAin}} \times A_{OL}(s)$$

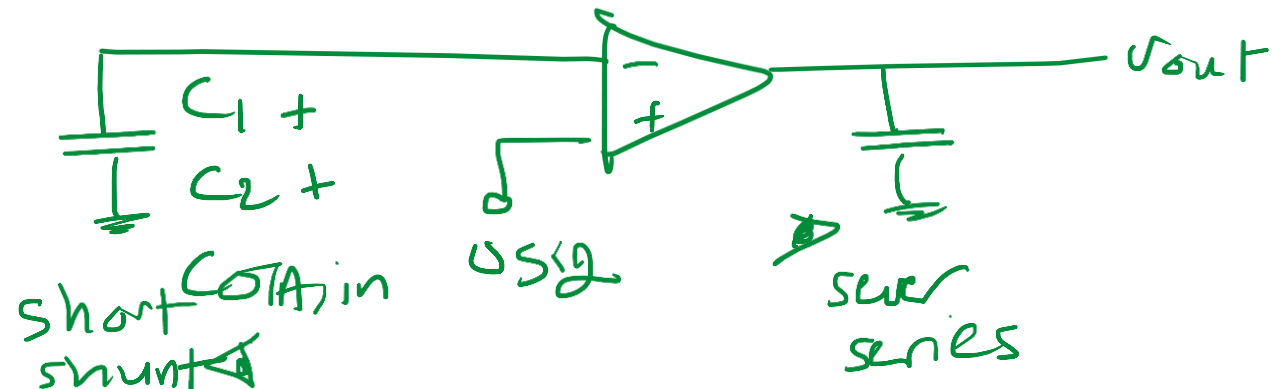
depends on C_{eff}



C_{eff} is correct here

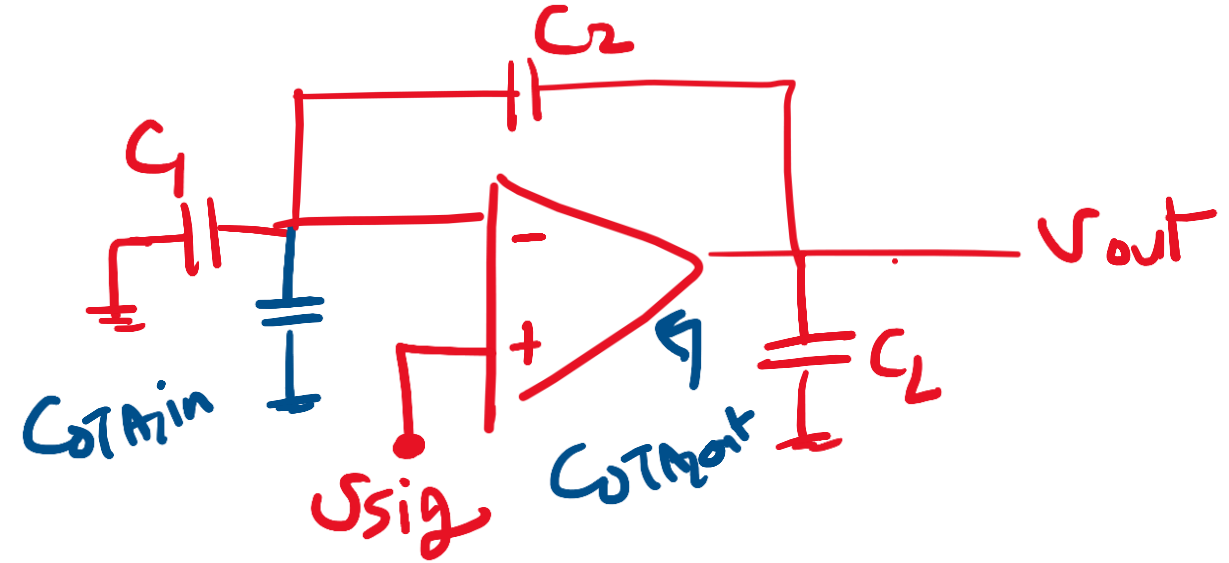
A-Circuit

$$C_{eff} = C_2 + C_{OTAout} + C_2 \parallel (C_1 + C_{OTAin})$$



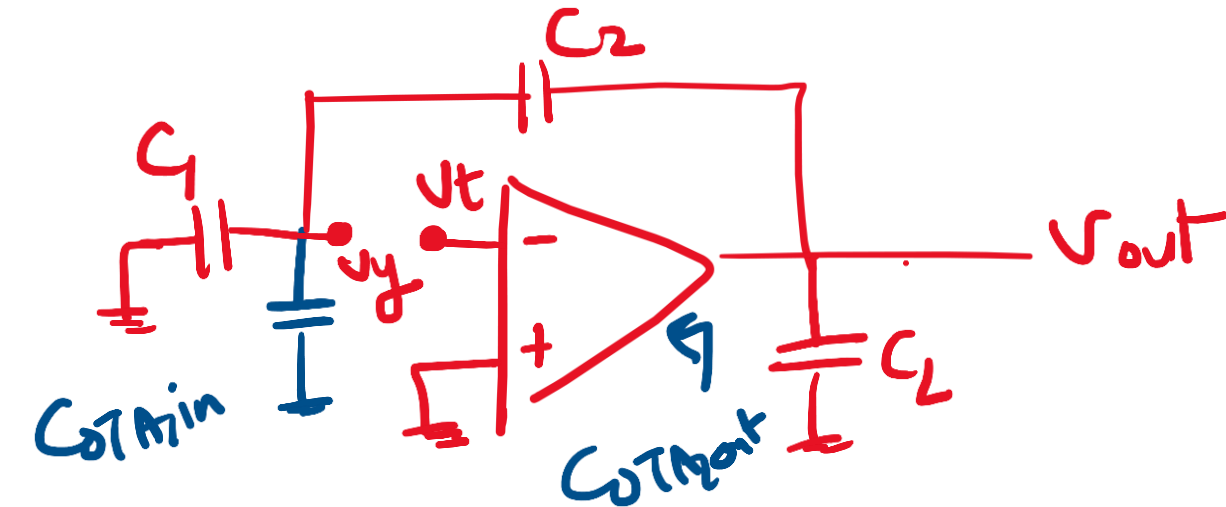
Non-Inverting (Series-Shunt) Analysis

$$\begin{aligned} A_{v,CL} &= \frac{v_{out}}{v_{sig}} \\ &\approx \frac{1}{\beta_V} \\ &= 1 + \frac{C_1 + C_{OTAin}}{C_2} \end{aligned}$$



Mapping Closed-Loop to Open-Loop Specs

- ❑ From CL static error (LGo) and beta -> Find OL Ao
- ❑ From the feedback network find CL,eff

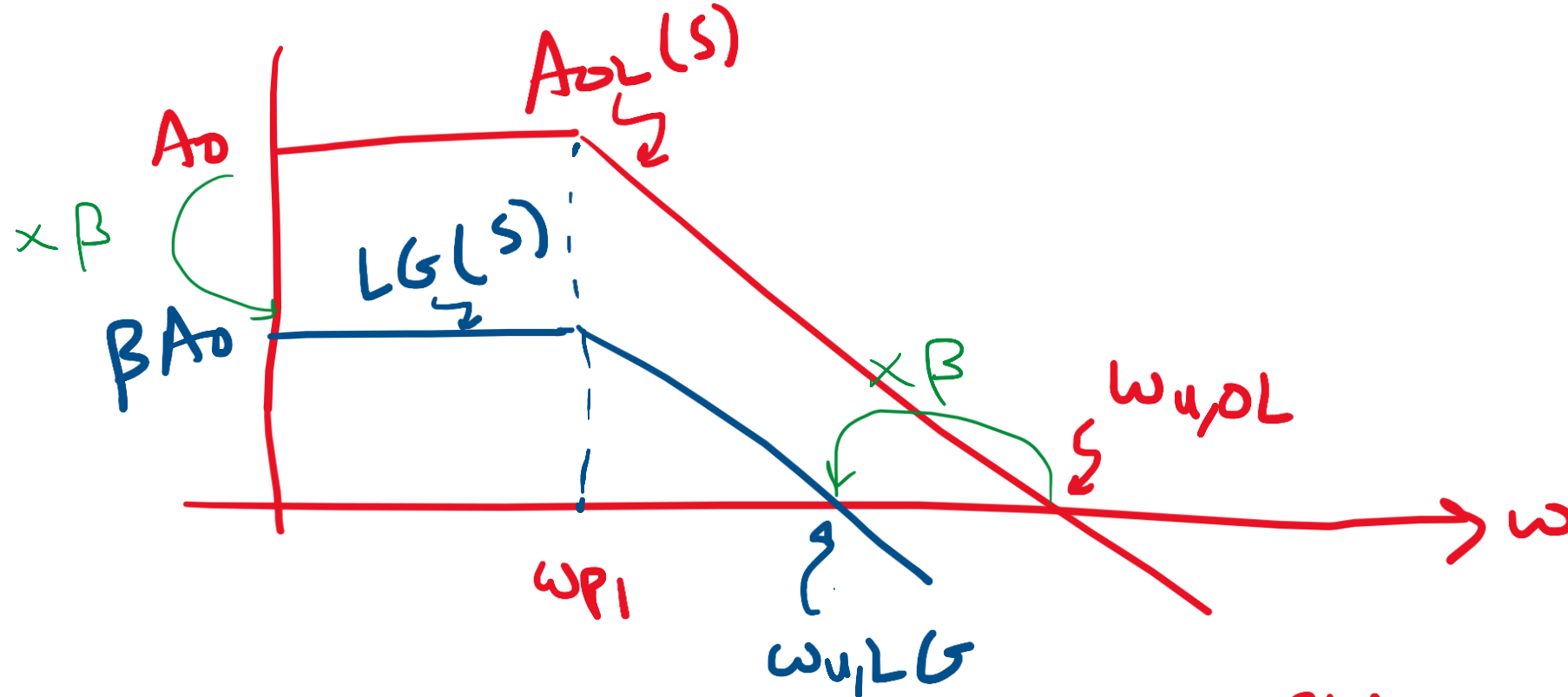


$$A_o = \frac{LG_o}{\beta}$$

$$C_{Leff} = C_L + C_{OTA,out} + C_2 \parallel (C_1 + C_{OTA,in})$$

Mapping Closed-Loop to Open-Loop Specs

- From CL BW (LG GX) find $\omega_{u,OL}$



$$\omega_{u,OL} = \frac{\omega_{u,CL}}{\beta} = A_v \times BW = \frac{BW_{CL}}{\beta}$$

Mapping Closed-Loop to Open-Loop Specs

- From PM,CL (PM,LG) find PM,OL

$$PM_x \approx 90^\circ - \tan^{-1} \left(\frac{\omega_{u,x}}{\omega_{p2}} \right)$$

(Note: ω_{p1} is written above the equation with an arrow pointing to the 90° term)

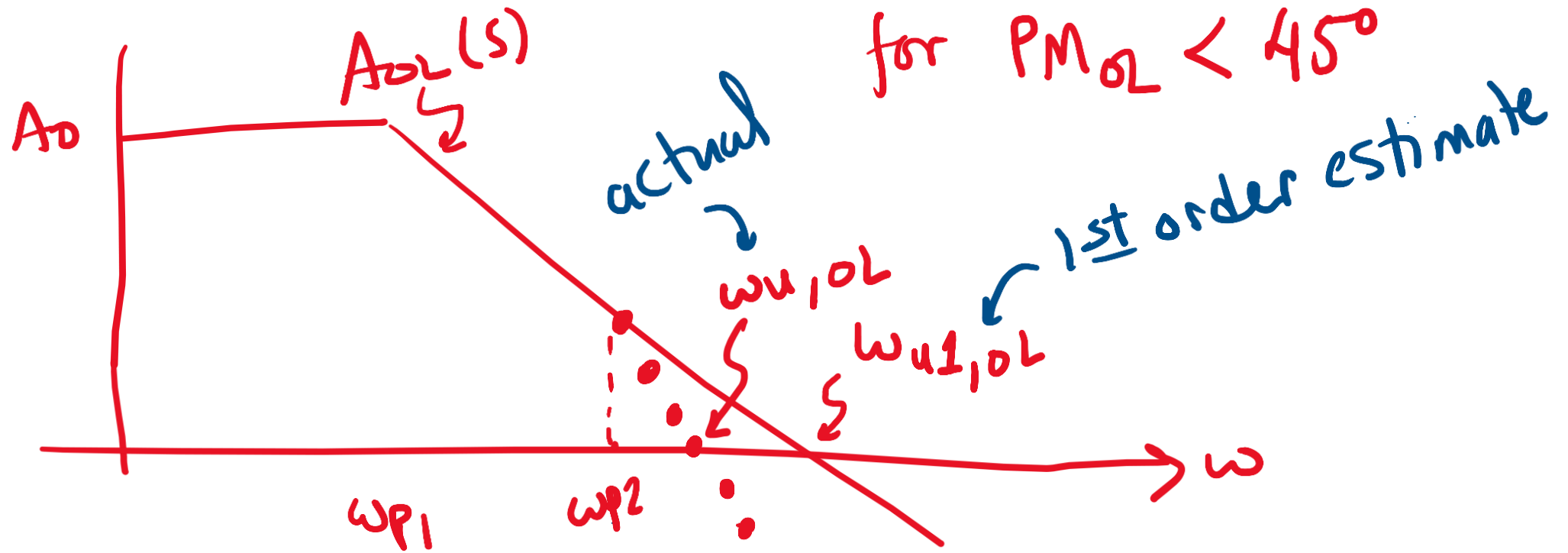
$$\omega_{u,OL} \rightarrow PM_{OL}$$

$$\omega_{u,LG} \rightarrow PM_{LG}$$

$$\text{Ex: let } PM_{LG} \sim 76^\circ \text{ \& } \beta = \frac{1}{3} \Rightarrow \frac{\omega_{p2}}{\omega_{u,LG}} \sim 3$$

$$\frac{\omega_{p2}}{\omega_{u,OL}} = \frac{\omega_{p2}}{\omega_{u,LG} / \beta} = 3 \times \beta = 1 \Rightarrow PM_{OL} \sim 45^\circ$$

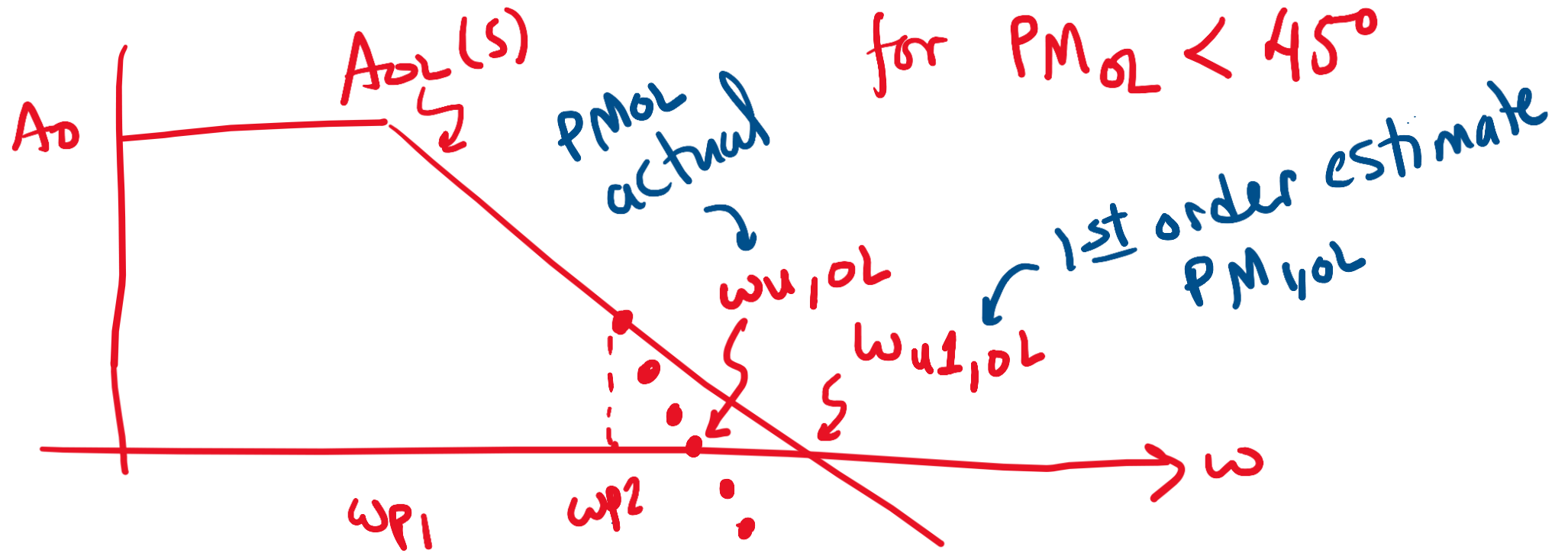
Designing Using Open-Loop Parameters



$$\omega_{u,OL} < \omega_{u1,OL} = \frac{BW_{CL}}{\beta}$$

If we design for $\omega_{u,OL} = \frac{BW_{CL}}{\beta} \rightarrow$ Actual BW_{CL} will be higher than expected

Designing Using Open-Loop Parameters



$$PM_{OL} > PM_{1,OL} = 90 - \tan^{-1} \left(\frac{\omega_{u1,OL}}{\omega_{p2}} \right)$$

If we design using PM_{OL} → Actual PM_{OL} will be lower than expected

Thank you!